PRIME'S MINISTER OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
SONGEA DISTRICT COUNCIL
FORM FOUR PRE – NECTA EXAMINATION

033/1 **BIOLOGY 1**

TIME: 3HOURS **JULY 2026**

INSTRUCTIONS.

1. This paper consists of section A, B and C with total of **eleven (11)** questions.
2. Answer all question in section A and B and **two** question from section C
3. Section A carries **16 marks** section B carries **54 marks** and section C carries **30 marks**
4. All writing should be in blue or black ink except for diagrams that must be draw in pencil
5. Remember to write your candidate examination number at the top of every page.
6. Cellular phones and any unauthorized materials are not allowed in the examination room

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SECTION A (16 marks)

Answer **all** questions in this section

1. For each of the items (i) - (x), choose the correct answer from among the given alternatives and its letter beside the item number in the answer booklet provided

(i) One of the following is not a characteristic of prokaryotic cell

(A) They have nuclear membrane
B) They reproduce by binary fission
(C) They have circular DNA
(D) They are small in size

(ii) Which of the following is a reason for studying biology?
(A) Biology help us to appreciate and improve the nature
(B) Biology deals with non-living thing objects in our environments
(C) Biology research contributes to the development of low-yielding breeds of crops

(D) Biology enables us to pollute the environments

(iii) A flowering plants with one embryonic seed cotyledon that usually appears at germination is;

(A) An apple
(B) Maize
(C) Bean
(D) Cashew nut

(iv) The mitotic stage in which the chromosomes are lined up at the equator of the spindle fibre is called;

(A) Prophase stage
(B) Interphase stage
(C) Metaphase stage
(D)Anaphase stage

(v) The set of observable characteristics that can be influenced by genetic and environmental factors is:

(A) Allele
(B) Genotype
(C) Phenotype
(D)Gene

(vi) Which of the following groups of animals are ectothermic?

(A) Bat, cow,bird and cat
(B) Dog, fish, alligator and frog
(C) Frog ,alligator, crocodile and fish
(D) Fish, man, frog, and crocodile

(vii) Crustaceans, arthropods and arachnids are made up of exoskeleton that limits growth in these animals. Which biological process do these animals use to overcome the problem of limited growth?

(A) Bursting
(B) Moulting
(C) Loss of body weight
(D) Ceasation

(Viii) Which of the two sets of hormones stimulates the production and ejection of milk from the nipples in lactating mothers;

(A) Trophic and growth hormones
(B) Antidiuretic and gonadal hormones

(C) Prolactin and oxytocin hormones (D) Adrenal and oxytocin hormones

(ix) Epiphytes are small plants that grow on big tree branches without causing harm to the big tree.Which form of symbiotic relationship is shown by epiphyte

(A) Mutualism (C) Saprophytism
(B) Parasitism (D) Commensalis

(x) The types of drugs that slow down the brain and makes a person feel sleepy is ;

(A) Stimulants (C) Hallucinogens
(B) Sedatives (D) Narcotics

2 .Match the terms in column **A** with that in column **B** by writing a correct letter of the corresponding item from column **B** against roman number in column **A**

Column A	**Column B**
(i) A type of germination where by cotyledons are pushed above the soil surface (ii) A pore which allow water to enter a seed during germination (iii) Inability of a seed to germinate (iv) Prevent a viable seed to germinate (v) It forms primary roots (vi) It give rise to a shoot	A. Micropyle B. Viability C. Cotyledon D. Epigeal germination E. Hard seed coat F. Radicle G. Dormancy H. Plumule I. Hypogeal germination

SECTION B (54marks)

Answer **all** question in this section

3. (a)Identify the field and branch of biology that you will use to solve the following problems

(i) Helping a person with a broken leg
(ii) Protecting cattle's from tsetse flies
(iii)Conserving village water catchments from drying

(b)With one example in each describe three main components of coordination in mammals

4. (a) During classroom teaching a biology teacher asked form four students to collect various organisms around school compounds. Which apparatus or equipment's they will use to perform each of the following activities;

(i) Collect insects and small animals from the school compound
(ii)Catch a frog from a pond in order to study its external features

(iii)Estimate the number of small organisms per square metre

(iv) Keeping specimens alive

(b)In a biology class, your fellow students had classified pigeon and butterflies in the same group based on the presence of wings. With **five points (5**) explain the shortcomings of their classification

5. (a) Explain in one sentence why the following features occurs in these organisms;

(i) Needle-like leaves of plants found in desert

(ii) Coiled and elongated ileum in humans

(iii) Ciliated trachea in human

(iv) Streamlined body of birds

(v) The moist surface in respiratory system

(b) Name the **type** of joint found at the articulation of:

i) Femur and tibia

ii) Humerus and ulna

iii) Humerus and scapula

iv) Femur and pelvic girdle

6. After a heavy rain season, a house of Mr Songambele was surrounded with stagnant water with green scum layer

a) Identify the micro-organisms that cause green scum on water

b) Draw a diagram of the micro-organism in **6**(a) above and label the parts with the following functions

i) Locomotion

ii) Regulating water content

iii) Photosynthesis process

(b) Growth in plants take place in different zones called meristems. Describe **three** types of meristems

7. A woman visited a food seller and bought the following foods; eggs; Irish potatoes and sunflower seeds

(a) Identify the type of nutrient found in each of the food items listed.

(b) Give four functions of the nutrients found in eggs

8. When a red-flowered rose plant was crossed with a white flowered rose plant all members of the F1 generation were pink. Using genetic diagrams illustrate the cross made

SECTION C (30 marks)

Answer only **two (2)** questions from this section

9. Analyze six contributions of genetics in our daily life

10. Mammals use both behavioral and physiological mechanisms in regulating their body temperature. Using **six points** explain how behavioral mechanism take place in mammals

11. You have been invited by your nearby village officer to talk about irresponsible sexual behavior. Explain six methods of reducing these behaviors that you would address to the villager